



Polar Bears and Lazy Lizards

Jared Haertel | Spring 2026

Field(s) of Interest: Biology, Engineering

Brief Overview (1-3 sentences):

Mentees will learn the scientific method in the context differentiating heat and temperature. They will also learn how mammals and reptiles stay warm through different adaptations.

Agenda:

- Introduction (5 min)
- Module 1: It's gettin hot in here (5-10 min)
- Module 2: Arm bathing (15-20 min)
- Module 3: Baby, it's cold outside (20-25 min)
- Conclusion (5 min)

Main Teaching Goals/Key Terms:

- Heat energy
- Temperature
- Control group
- First law of thermodynamics
- Second law of thermodynamics
- Endotherm and ectotherm (or warm- and cold-blooded)
- Adaptations
- Conductors
- Insulators
- The scientific method
 - ◆ Question
 - ◆ Hypothesize
 - ◆ Experiment

Background for Mentors

Module 1 • Heat energy • Temperature • Control group • First law of thermodynamics • Second law of thermodynamics	Temperature measures the average kinetic energy of an object. In everyday life, we measure heat with thermometers, which work because the liquid inside (traditionally mercury) expands with temperature. Heat energy is the form of energy that <i>crosses</i> the boundary as a result of differing temperatures. When you touch an object, you are not necessarily measuring the temperature, but rather the heat flow between your body and the object. The reason why a metal spoon feels colder than a plastic spoon which are both at room temperature is because the metal spoon absorbs heat faster. We will elaborate on this in the next module. In an experiment, scientists (or engineers) try some sort of treatment or solution and want to prove to others that they were successful. A control group is a group which forms a baseline, allowing scientists or engineers to determine the level of change compared to the group that receives a treatment (or in the case of engineering, their proposed solution). It also helps account for environmental variables, such as ambient temperature in the case of this lesson. In a clinical setting, a good control group would have people of a similar demographic as the treatment setting. For instance, it would be unfair to evaluate how fit elderly people are after not receiving a fitness plan to young people who do receive a fitness plan from the researchers. The first law of thermodynamics states that energy cannot be created nor destroyed, but only transferred and converted to other forms of energy. A light bulb turns electricity to light and pushing a ball from a hill converts potential energy to kinetic energy. The second law of thermodynamics states that the total entropy (or disorder) of an isolated system cannot decrease. For the purposes of our lesson, this implies that heat spontaneously flows from hotter to colder bodies. This can only be reversed with external work, such as cooling pumps in a refrigerator.
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Module 2

- Endotherm and ectotherm (or warm- and cold-blooded)
- Adaptations
- Conductors
- Insulators

Adaptations are traits that a species has evolved to help increase their chance of survival in an environment. An example would be humans being bipedal to walk great distances.

Endotherms (or **warm-blooded**) organisms are those that can maintain their body temperature to support their necessary biological processes (i.e. maintain metabolism). Birds and mammals are considered the only endotherms. Some adaptations include fat, fur or feathers, shivering when cold, or burning brown fat when cold. Humans have adapted to sweating to This, however, requires a lot of energy. For humans, 40-50% of our calories go to maintaining our own body temperature.

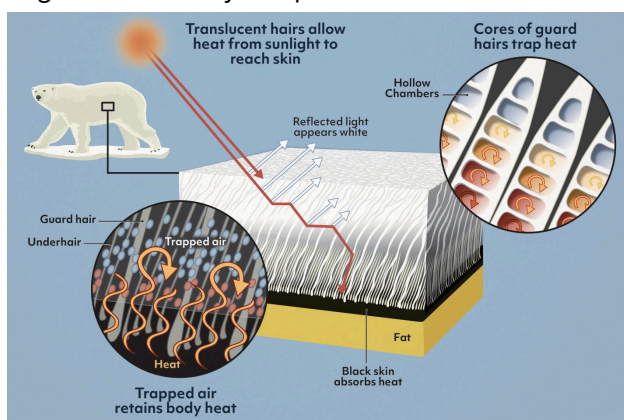


Figure 1: Strategies polar bears use to keep warm

Ectotherms (or **cold-blooded**) organisms are those that rely on the environment to stay at metabolic temperatures. Reptiles are ectotherms and often bask in the sun and lie on warm rocks to heat up, or hide in shade to cool down. This is a much more energy efficient adaptation, as they don't need to regulate their own heat!

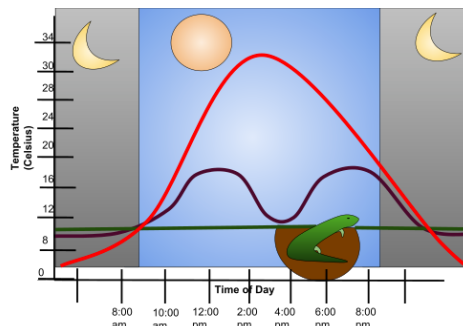


Figure 2: Ambient temperature (red) versus lizard body temperature (black) throughout the day. The green line temperature

The conductance of an object measures how fast heat flows for a given

temperature difference. The conductance k can be expressed in the equation

$$\mathbf{q} = -k\nabla T$$

where $\mathbf{q}$ is the heat flux, and ∇T is the temperature gradient. A higher conductance corresponds to a (thermal) **conductor**, such as metals or water, which can quickly transfer heat. A lower conductance correspond to a (thermal) **insulator**, such as plastic or foam, which resists the flow of heat.

Module 3

- The scientific method
 - Observe
 - Question
 - Research
 - Hypothesize
 - Experiment
 - Analyze results

The **scientific method** is an empirical way of gaining knowledge through observation, careful scrutiny, hypothesis testing and replicating experiments. It can be very powerful to answer questions about the world (“what”, “how”, “can”, “when”, “why”, “where”), but cannot answer questions of morality (“should”). It can be roughly decomposed into a few stages, where we use the example of engineering a thermos

The **observation** stage involves making note of things in the world that then lead to **questions**. Science stems from humans' curiosity about the world.

The **research** stage involves gathering background information about that topic to make a well-informed guess, or **hypothesis**.

From this guess, you design a repeatable, falsifiable **experiment** that can prove or disprove your claim. It is then important to **analyze the results**, to demonstrate that the experiment shows what you claim it does and evaluate your hypothesis.

These experiments then add to your observations to form new questions or revisions to your hypothesis, leading to more science.

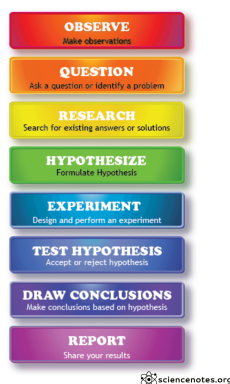


Figure 1: The scientific method

As an example from Crash Course Biology, one can observe that an egg has a lot of water, and you wonder what would happen if you microwave it. You perhaps read that a microwave heats up water molecules, and they will turn to steam, putting pressure on the eggshell from within. So you hypothesize that the egg will explode! To test this, you microwave an egg for a minute and see what happens. After it blows up, you then hypothesize that breaking the eggshell will allow steam to escape meaning it won't blow up, leading to another hypothesis.

Introduction

Concepts to Introduce • Have mentees touch different things in a classroom, such as their wooden table or metal on their chair. These are actually the same temperature! What does it mean for things to be hot or cold? • The scientific method allows for precise claims to be proven or disproven, pushing the boundary of knowledge forward.	Questions to Pique Interest • What percent of the energy from the food you eat goes to keeping you warm? ◦ Shockingly, it's 40-50%! • Why do lizards live in warmer environments while mammals dominate colder regions? ◦ Mammals can produce their own heat but lizards rely on the heat from around them.
Scientists, Current and Past Events • Thomas Newcomen invented the steam engine in 1712, which could turn the heat energy from steam into motion. • Computer datacenters require lots of water and energy to keep cool. This has raised concerns on their environmental impact.	Careers and Applications • Geothermal engineers use heat from the hot water deep in the ground to create energy. • Computer engineers need to use conductors to cool down their chips effectively. • MRI machines use both conductors and insulators to keep the machine near absolute zero (around -454 F)! • Biologists study how different organisms can thrive in very different climates.

Module 1: It's gettin hot in here

Mentees will learn the difference between temperature and heat by doing a small group demo to see how materials affect heat conductance.

Teaching Goals 1. Heat energy: The form of energy that crosses a boundary of an object due to a temperature difference 2. Temperature: Proportional to the average kinetic energy of an object 3. Control group: A baseline for comparison in an experiment 4. First law of thermodynamics: energy cannot be created nor destroyed, only transferred and converted 5. Second law of thermodynamics: Heat cannot spontaneously flow from cold to hot.	Materials for demo • Spoon • Coconut oil tub • Warm water in thermos • Bowls
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Different Methods for Teaching

1. **Heat versus temperature:** Temperature is the **total** amount of thermal energy while heat is the **transfer** of thermal energy. That is why when you touch something, you are “measuring” heat, not temperature. To measure temperature you need a thermometer!
2. **Control group:** Throughout each module, there is some sort of control group. For instance, in module 1 the plastic spoon could be the control. In module 2, it could be the paper lizard, and in module 3 it could be the stand alone paper cup. Explain to mentees how comparing to a control group helps scientists understand the effect of their changes compared to the control.
3. **Second law of thermodynamics:** Heat is like a river flowing from a mountain, it naturally wants to go hot to cold. But we can use a pump to reverse the direction of travel such as in a fridge.

Procedure for the demo

1. Form small groups with one mentor per group. Have mentees touch both spoons and say which one they think is colder.
2. Put a little coconut oil on both spoons and have mentees carefully put the bottom of the spoon in a container of warm water
3. Which melts first?
4. **Put water back into thermos** (so that you have enough for module 3)
 - a. It's okay if there is oil in the water



Figure 3: The setup

Classroom Notes

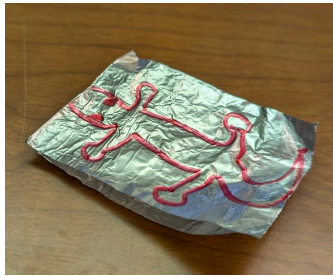
It may be hard to see, so have mentees crowd around the mentors. It's important that water doesn't spill in so take your time to aim!

Module 2: Arm bathing

Mentees will individually build an aluminum lizard and paper lizard and see which is better at soaking up heat from the environment (or in this case, the mentees arms). The aluminum and paper are examples of conductors and insulators respectively.

Teaching Goals 1. Endotherm and ectotherm (or warm- and cold-blooded): Organism that primarily produces their own heat versus organism that relies on ambient heat 2. Adaptations: Traits that help the survival of a species 3. Conductors: Objects that have fast heat transfer 4. Insulators: Objects that have slow heat transfer	Materials (Per mentee) 1. 1 aluminum strip 2. 1 index card 3. 2 thermochromic rectangles 4. 2 glue dots 5. Scissors 6. Markers
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Different Methods for Teaching 1. Endo what?: For younger sites, you can just use the terms warm- and cold-blooded instead of endotherm and ectotherm. As a shorthand, “endo” sounds like “indoor”, so “indoor-therm” means heat is made from inside! 2. Adaptations: Emphasize that being an endotherm requires a lot of energy (and thus food) which may make sense in colder environments. Lizards tend to live in hot environments, where one can save on energy by using the environment instead. 3. Conductance: Talk about how conductors and insulators are used in everyday life! Cooking ware tends to be made out of metal (a conductor) to spread out the heat evenly to cook food. The fluff in coats keeps the body warm and fiberglass insulation keeps one's home warm.

Procedure 1. Fold the aluminum strip in half 2. On both, draw a lizard of equal size on both, but require that the head has a flat edge along the fold for the aluminum foil (see figure 1) so that it stays in one piece when cut 3. Cut out the lizards! a. It's easiest to start from the tail and work one's way up 4. Pinch or fold the feet, legs, tail and head to make more lizard like 5. Put a glue dot on the “back” of the lizards 6. Place thermochromatic rectangle over the dot	 Figure 1: Example lizard design
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7. Set them on the table until both rectangles are black
8. **If it's warm/hot in your room, you can either:**
 - a. Pinch the heads of both lizards and see which one has heat travel up it faster and further.
 - b. Warm up both lizards evenly on your arm. Then place them both on the table to see which cools down the fastest.
9. **If it's a cold day (sweater weather):**
 - a. Place both lizards on one's forearm, pushing on the edges to match the curve of the arm for better contact.
10. If time allows, decorate!



Figure 2: Easier design for younger mentees

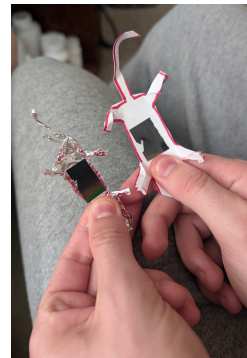


Figure 3: Pressing the heads. Note the heat travels further up!



Figure 4: Arm bathing

Classroom Notes

If it's not working that well, warm them both up to be the same color, and see which cools off faster.

Module 3: Baby, it's cold outside

In groups, mentees will build a thermos to model how polar bears stay warm. They will apply principles from the scientific method to verify their thermos works.

Teaching Goals 1. The scientific method a. Observe: Make observations b. Question: Form a question or identify a problem c. Research: Search for existing answers or solutions d. Hypothesize: Form a specific educated guess e. Experiment: Formulate an experiment that will prove or disprove your guess f. Analyze results: Does your experiment prove or disprove your guess	Materials (Per group) 1. Two small cups 2. Larger red cup 3. Cotton balls 4. Misc. materials a. Felt strips b. Pom poms c. Scrap paper 5. Tape 6. Scissors 7. Thermometers 8. Water in thermos
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Different Methods for Teaching

- 1. So many teaching goals!** The most important teaching goals are to question, hypothesize, and form a good experiment, so emphasize these during the activity!
- 2. Tie it back to the last modules teaching goals:** Why might it be beneficial for polar bears to be endotherms? How are they able to stay warm in the arctic? Are these examples of conductors or insulators?
- 3. Experimental design:** This can get tricky especially for younger mentees. You can have mentees imagine they are in a court of law, and they need to prove beyond reasonable doubt that their solution works. Some guiding questions could be:
 - a. How can you measure “warmness”?
 - b. How much time should there be between measurements?
 - c. Is their experiment “fair” to both sides? For example, the time between temperature readings and the starting temperature should be the about the same

Procedure

1. Form groups of around 4-5 mentees
2. With possible insulators laid out, have mentees decide which ones they want to use and have them explain why.
3. Have them design an experiment to prove that insulators can keep a cup of water warmer for longer. **Have them come up with a control group.**
4. As a general guide, a thermos can be built as follows:
 - a. Put a little bit of any insulator at the bottom of the red solo cup
 - b. Put the smaller paper cup so that it is centered
 - c. Fill the sides with more insulator
 - d. Make some form of a lid
5. Once built, fill the thermos less than a third of the way up, and do the same for the spare paper cup which will act as a control
 - a. **Note:** If your site is larger (more than ~25 mentees) you can either:
 - Put less water in each cup
 - Bring more hot water to the site with your own thermos
 - Mix a little bit of tap water
6. Measure the temperature of the water over time!
 - a. For younger sites, maybe only record the starting temperature and the final temperature after some time
 - b. For advanced sites, maybe record temperature at multiple points
7. Ask mentees if the results align with their predictions.



Figure 1: Example prototype



Figure 2: Example cross-section



Figure 3: Experiment in progress

Classroom Notes

For the lid, mentees can just fold a few layers of paper over the top.

Conclusion

Ask mentees if they prefer warming up by laying in hot sand like a lizard or by burying themselves in blankets like a polar bear.

Summary Materials Table

Material	Amount per Site	Expected \$\$	Vendor (or online link)
Metal spoon	1 per mentor	\$27	Amazon
Plastic spoon	1 per mentor	In MCL	
Coconut oil	A tub per site	\$54	Amazon
Thermos	1 per site	In MCL	
Bowls	1 per mentor	In MCL	
Aluminum foil	1 strip per mentee	In MCL	
Index card	1 per mentee	\$8	Amazon
Thermochromatic sheets	2 rectangle per mentees	\$53	Amazon
Glue dots	2 per mentee	In MCL	
Scissors	Few per site	In MCL	
Markers	Few per site	In MCL	
Small paper cups	1 per group	\$22	Amazon
Larger cups	1 per group	In MCL	
Cotton balls	Handful per group	\$29	Amazon
Fabric sheets	Few per site	In MCL	
Pom poms	Few per site	In MCL	
Tape	2 rolls per site	In MCL	
Thermometers	1 per group	\$31	Amazon